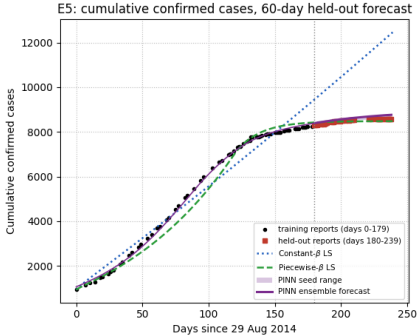
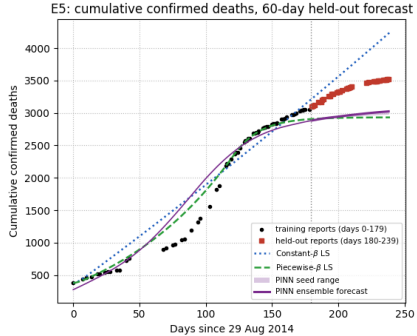
E5: Held-out forecast on WHO data

Module 5 ▪ WHO lab-confirmed reports, Sierra Leone
fit: 29 Aug 2014 – 24 Feb 2015 ▪ test: 25 Feb – 25 Apr 2015, never seen





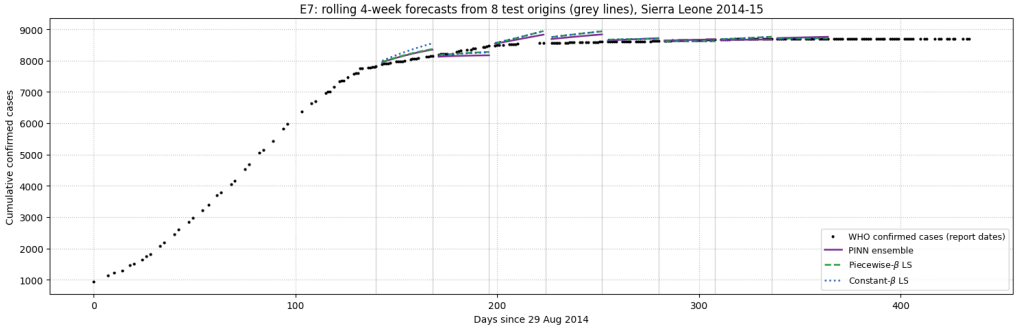
Cases, relative error	PINN	Piecewise-β	Constant-β
60-day forecast	1.35%	**0.82%**	28.6%

Deaths, relative error	PINN	Piecewise-β	Constant-β
60-day forecast	11.8%	12.9%	**10.7%**

A time-varying β is what matters: Constant-β keeps growing and misses cases by 29%. **PINN** is a close second to Piecewise-β on cases (single seeds: 0.28–1.75%). **Deaths defeat every method** (11–13%): WHO added 463 deaths but only 361 cases in the test window.

E7: Rolling 4-week forecasts across the epidemic

Module 5 ▪ 8 origins, every 28 days from 16 Jan to 31 Jul 2015 ▪ each fit sees only the previous 8 weeks



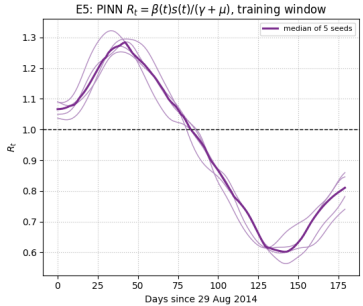
Mean of 8 origins	Cases MAE	New cases, 4 wk	Deaths MAE
PINN (3 seeds)	**96.8**	**148.2**	69.7
Piecewise-β	106.5	167.8	**42.0**
Constant-β	120.9	192.5	50.7

Errors in people. "New cases, 4 wk": error in new cases over the horizon.

✓ **Cases:** **PINN** is best on average, 20% below **Constant-β**, and beats **Piecewise-β** at 6 of 8 origins.

× **Deaths:** **PINN** is worst; its death forecast inherits any error in the learned μ.

What the PINN adds, and what it costs



Learned $R_t = \beta(t)\, s(t)/(\gamma+\mu)$, E5 training window (5 seeds, median bold):

- peak ≈ 1.28 in early Oct 2014
- crosses 1 in mid-Nov 2014: the epidemic turns
- ≈ 0.6 in mid-Jan 2015, ≈ 0.8 by late Feb 2015

What it gives that least squares does not

- a continuous $R_t(t)$ with no breakpoints to choose
- the case-fatality ratio, learned: 0.347–0.357 across seeds
- the hidden compartments S, E, I alongside the fit

What it costs

- 1–2 min per fit (Colab CPU) vs. seconds for least squares
- needs a seed ensemble: single seeds vary (0.28–1.75%)
- forecasts hold β at its recent value, so they cannot foresee a new intervention or a resurgence

Honest verdict: on cases the PINN forecasts about as well as the best classical fit; on deaths it does worse. Its real value is an interpretable, continuous picture of transmission.